Since n is large and p is small, Binomial Distribtuion approximated to poisson distribution

So, $\lambda = np = 2000 \times 0.001 = 2$

(i) Probability out of 2000, exactly 3 will suffer a bad reaction is

$$P(X = 3) = \frac{e^{-\lambda}\lambda^{x}}{x!} = \frac{e^{-2}2^{3}}{3!} = 0.1804$$

(ii) Probability out of 2000, more than 2 individuals will suffer a bad reaction

$= P(X > 2)$

$= 1 - [(P(X \leq 2)]$

$= 1 - [P(x = 0) + P(x = 1) + P(x = 2)]$

$$= 1 - \left[\frac{e^{-2}2^{0}}{0!} + \frac{e^{-2}2^{1}}{1!} + \frac{e^{-2}2^{2}}{2!}\right]$$

$$= 1 - e^{-2}\left(\frac{2^{0}}{0!} + \frac{2^{1}}{1!} + \frac{2^{2}}{2!}\right)$$

$= 0.323$

Example 7.19

When counting red blood cells, a square grid is used, over which a drop of blood is evenly distributed. Under the microscope an average of 8 erythrocytes are observed per single square. What is the probability that exactly 5 erythrocytes are found in one square?

Solution :

Let X be a random variable follows poisson distribution with number of erythrocytes.

Hence, Mean $\lambda = 8$ erythrocytes/single square

P(exactly 5 erythrocytes are in one square)

$$= P(X = 5) = \frac{e^{-\lambda}\lambda^{x}}{x!} = \frac{e^{-8}8^{5}}{5!}$$

$$= \frac{0.000335 \times 32768}{120}$$

$= 0.0916$

The probability that exactly 5 erythrocytes are found in one square is 0.0916. i.e there are 9.16% chances that exactly 5 erythrocytes are found in one square.

Example 7.20

Assuming one in 80 births is a case of twins, calculate the probability of 2 or more sets of twins on a day when 30 births occur.

Solution :

Let x denotes the set of twins on a day

P(twin birth) $= p = 1/80 = 0.0125$ and $n = 30$

The value of mean $\lambda = np = 30 \times 0.0125 = 0.375$

Hence, X follows poisson distribution with

$$p(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

The probability is

$P(2 \text{ or more}) = 1 - [p(x = 0) + p(x = 1)]$

$$= 1 - [\frac{e^{-0.375}(0.375)^{0}}{0!} + \frac{e^{-0.375}(0.375)^{1}}{1!}]$$

$= 1 - e^{-0.375}\ [1+0.375]$

$= 1 - (0.6873 \times 1.375)$

$= 0.055$

Exercise 7.2

1. Define Poisson distribution.
2. Write any 2 examples for Poisson distribution.
3. Write the conditions for which the poisson distribution is a limiting case of binomial distribution.
4. Derive the mean and variance of poisson distribution.
5. Mention the properties of poisson distribution.
6. The mortality rate for a certain disease is 7 in 1000. What is the probability for just 2 deaths on account of this disease in a group of 400? [Given $e^{(-2.8)} = 0.06$]

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